

DBMS Interview Questions

1. What is DBMS?

A DBMS, or Database Management System, is software used to store, retrieve, and manage data efficiently. It allows multiple users to interact with data through structured queries.

2. What's the difference between DBMS and RDBMS?

RDBMS is an advanced form of DBMS that stores data in tabular form and supports relationships using keys. Examples include MySQL, PostgreSQL, and Oracle. DBMS doesn't enforce relationships, like in file systems.

3. What is a Primary Key?

A primary key uniquely identifies each record in a table. It cannot have `NULL` or duplicate values.

4. What is a Foreign Key?

A foreign key links two tables together. It refers to the primary key of another table, enforcing referential integrity.

5. What is a Candidate Key?

Candidate keys are all possible keys that can uniquely identify a record. Out of them, one is chosen as the primary key.

6. What is a Composite Key?

A composite key is made up of two or more columns that together uniquely identify a record.

7. What are Constraints in SQL?

Constraints are rules applied to columns to maintain data integrity — examples include `PRIMARY KEY`, `FOREIGN KEY`, `UNIQUE`, `CHECK`, and `NOT NULL`.

8. Explain the difference between DELETE, TRUNCATE, and DROP.

- `DELETE` removes selected rows and can be rolled back.
- `TRUNCATE` removes all rows quickly and cannot be rolled back.

- **DROP** removes the entire table structure.
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9. What is Normalization?

Normalization is the process of organizing data to reduce redundancy and improve integrity by dividing data into related tables.

10. Explain 1NF, 2NF, and 3NF.

- **1NF:** Each cell holds a single value.
 - **2NF:** No partial dependency on a composite key.
 - **3NF:** No transitive dependency (non-key depends on non-key).
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11. What is Denormalization?

Denormalization is the process of combining normalized tables to improve read performance, at the cost of redundancy.

12. What is a View?

A view is a virtual table created using a SQL query. It doesn't store data but displays data from underlying tables.

13. What are Joins?

Joins combine rows from two or more tables based on a related column.

14. Name different types of Joins.

- Inner Join
 - Left Join
 - Right Join
 - Full Outer Join
 - Cross Join
 - Self Join
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15. What is an Inner Join?

An inner join returns only those records that have matching values in both tables.

16. What is a Self Join?

A self join is when a table is joined with itself to compare rows within the same table.

17. What is a Cartesian Join?

It returns the Cartesian product — every row of one table combined with every row of another table.

18. What is a Subquery?

A subquery is a query inside another query, used to fetch data for comparison or filtering.

19. What are Correlated Subqueries?

A correlated subquery depends on the outer query — it's executed once per row of the main query.

20. What is an Index?

An index improves the speed of data retrieval operations on a table. It acts like a pointer for quick lookups.

21. What are the types of Indexes?

Common types include **Clustered**, **Non-Clustered**, **Unique**, and **Composite Indexes**.

22. What is a Clustered Index?

A clustered index sorts and stores the rows physically in the order of the key values. Each table can have only one clustered index.

23. What is a Non-Clustered Index?

A non-clustered index creates a separate structure for the index that points to the actual table rows.

24. What is a Transaction?

A transaction is a single logical unit of work that must either be completed entirely or not at all.

25. Explain ACID Properties.

- **Atomicity:** All or none.

- **Consistency:** Maintains data integrity.
 - **Isolation:** Transactions don't interfere.
 - **Durability:** Once committed, changes persist.
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26. What is a Deadlock?

A deadlock occurs when two transactions hold locks and wait for each other indefinitely.

27. How can you prevent Deadlocks?

By acquiring locks in a fixed order, using shorter transactions, and keeping consistent access patterns.

28. What are Stored Procedures?

Stored procedures are precompiled SQL code stored in the database for reuse and performance improvement.

29. What is a Trigger?

A trigger is a stored procedure that automatically executes when a specified event (INSERT, UPDATE, DELETE) occurs on a table.

30. Difference between Stored Procedure and Trigger?

Stored procedures are manually invoked, while triggers fire automatically on specific events.

31. What is a Cursor?

A cursor allows row-by-row processing of query results — used when individual record handling is needed.

32. What are Transactions Isolation Levels?

They control concurrency problems. Common levels:

- Read Uncommitted
 - Read Committed
 - Repeatable Read
 - Serializable
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33. What are the anomalies prevented by Normalization?

- **Insertion anomaly**
 - **Updation anomaly**
 - **Deletion anomaly**
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34. What is Referential Integrity?

It ensures relationships between tables remain consistent — foreign key values must match existing primary keys.

35. What is a Schema?

A schema is the logical structure that defines how data is organized — includes tables, views, indexes, etc.

36. What is the difference between Schema and Instance?

Schema is the blueprint of the database (static), while instance is the actual data at a given moment (dynamic).

37. What is the difference between DBMS and File System?

DBMS provides efficient data management with indexing, transactions, and security — unlike file systems, which lack integrity and relational features.

38. What is an Entity?

An entity is a real-world object or concept represented in a database — like `Student` , `Course` , or `Employee` .

39. What is an Attribute?

An attribute is a property or characteristic of an entity, such as `Name` , `Age` , or `Salary` .

40. What is an Entity-Relationship (ER) Diagram?

An ER diagram visually represents entities, attributes, and their relationships in a database design.

41. Explain 1:1, 1:N, and M:N relationships.

- **1:1**: One entity relates to exactly one other.
 - **1:N**: One entity relates to many others.
 - **M:N**: Many entities relate to many others.
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42. What is Data Redundancy?

It's the unnecessary duplication of data across tables, which normalization helps eliminate.

43. What is Data Integrity?

It refers to maintaining accuracy and consistency of data throughout its lifecycle.

44. What is a Tuple?

A tuple represents a single row in a table.

45. What is the difference between DDL and DML?

- **DDL (Data Definition Language):** Defines structure — `CREATE` , `ALTER` , `DROP` .
 - **DML (Data Manipulation Language):** Handles data — `SELECT` , `INSERT` , `UPDATE` , `DELETE` .
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46. What are DCL and TCL?

- **DCL:** Controls access (`GRANT` , `REVOKE`).
 - **TCL:** Manages transactions (`COMMIT` , `ROLLBACK` , `SAVEPOINT`).
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47. What is a Check Constraint?

It enforces a condition on a column to accept only values satisfying the rule, e.g. `CHECK (salary > 0)` .

48. What is the difference between HAVING and WHERE?

`WHERE` filters rows before grouping, while `HAVING` filters groups after aggregation.

49. Explain the difference between UNION and UNION ALL.

`UNION` removes duplicates, while `UNION ALL` includes duplicates and is faster.

50. What is a Materialized View?

A materialized view stores the result of a query physically for faster access, unlike a regular view which is virtual.

51. What is a Database Instance?

A database instance is the set of memory structures and background processes that manage database files. It's a running environment of a database.

52. What is meant by Data Independence?

Data independence allows you to change the schema at one level without affecting other levels — achieved through abstraction in DBMS.

53. What are the two types of Data Independence?

- **Logical Data Independence:** Changes in logical schema don't affect application programs.
 - **Physical Data Independence:** Changes in physical storage don't affect logical structure.
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54. What is a Data Dictionary?

It's a metadata repository storing information about database objects — tables, columns, users, constraints, etc.

55. What is the difference between Static and Dynamic SQL?

Static SQL is fixed at compile-time. Dynamic SQL is built and executed at runtime (like in stored procedures).

56. What is the difference between UNION and JOIN?

UNION combines rows vertically from different tables, while **JOIN** combines data horizontally based on related columns.

57. Explain what a NULL value represents.

NULL represents missing or unknown data — it's not zero or empty string.

58. How does SQL handle NULL in comparisons?

Any comparison with NULL returns NULL. To check, use **IS NULL** or **IS NOT NULL**.

59. What is the difference between ISNULL() and COALESCE()?

ISNULL(expr, value) replaces NULL with value.

COALESCE(expr1, expr2, ...) returns the first non-NULL value among given expressions.

60. What is the difference between a Function and a Procedure?

Functions return a single value and can be used in SELECT statements, whereas procedures may return multiple values and are invoked independently.

61. What is a Trigger used for in real scenarios?

Used for enforcing business rules, logging changes, or automatic auditing after insert/update/delete actions.

62. What is a Savepoint?

A savepoint is a checkpoint within a transaction that allows partial rollback instead of rolling back the entire transaction.

63. What is the purpose of COMMIT and ROLLBACK?

`COMMIT` permanently saves changes; `ROLLBACK` undoes them.

64. What is a Lock?

A lock is a mechanism to control concurrent access to data and ensure transaction isolation.

65. What are Shared and Exclusive Locks?

- **Shared Lock:** Allows read access, no modification.
 - **Exclusive Lock:** Allows data modification, no other access.
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66. What is Phantom Read?

It's a concurrency issue where new rows are added by another transaction between two reads of the same query.

67. What is Dirty Read?

Reading uncommitted changes from another transaction — may lead to inconsistent results.

68. What is Non-Repeatable Read?

When a row is read twice and yields different values because another transaction updated it in between.

69. What is a Database Cursor used for?

Used for iterating through result sets row by row when set-based operations aren't sufficient.

70. What is Query Optimization?

Query optimization is the process of choosing the most efficient query execution plan by the DBMS query optimizer.

71. What factors affect query performance?

Indexes, table size, join type, data distribution, and query structure.

72. How does an Index improve performance?

It reduces the number of data pages scanned by using a sorted data structure like a B-tree for quick lookups.

73. What is the drawback of too many indexes?

They speed up reads but slow down writes (INSERT/UPDATE/DELETE) because indexes must also be updated.

74. What is a B-tree Index?

It's a balanced tree structure that stores sorted keys for fast retrieval and range searches.

75. What is a Hash Index?

It uses a hash function to map keys to fixed buckets — fast for equality searches but not for range queries.

76. What is the difference between OLTP and OLAP?

- **OLTP (Online Transaction Processing):** For day-to-day transactions (frequent reads/writes).
 - **OLAP (Online Analytical Processing):** For complex analytical queries and data aggregation.
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77. What is Data Warehousing?

A data warehouse stores large volumes of historical data for analysis and reporting purposes.

78. What is a Distributed Database?

A distributed database stores data across multiple physical locations but appears as a single logical database to users.

79. What is Horizontal vs Vertical Partitioning?

- **Horizontal:** Splitting rows into multiple tables (e.g., by region).
 - **Vertical:** Splitting columns into multiple tables (e.g., separate frequently accessed columns).
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80. What is the difference between Logical and Physical Data Models?

The logical model defines entities, relationships, and constraints. The physical model defines storage, indexing, and partitioning.

81. What is E-R Modeling?

Entity-Relationship modeling is a high-level conceptual data model used during database design to identify entities and their relationships.

82. What is Weak Entity?

An entity that cannot be uniquely identified without a foreign key from another entity.

83. What is a Derived Attribute?

An attribute derived from other attributes — e.g., `Age` derived from `DateOfBirth`.

84. What is Data Abstraction in DBMS?

It hides complexity by organizing data into three levels: physical, logical, and view levels.

85. What are Integrity Constraints?

Rules that maintain correctness of data — includes domain, entity, and referential integrity.

86. What is Domain Integrity?

Ensures values in a column fall within a valid range or data type.

87. What is Functional Dependency?

A relationship where one attribute uniquely determines another — denoted as $X \rightarrow Y$.

88. What is Transitive Dependency?

When $A \rightarrow B$ and $B \rightarrow C$, then $A \rightarrow C$ is a transitive dependency.

89. What is BCNF (Boyce-Codd Normal Form)?

A table is in BCNF if every determinant is a candidate key — it removes all anomalies left in 3NF.

90. What are Anomalies in Databases?

They are inconsistencies or redundancies in unnormalized data — insertion, update, and deletion anomalies.

91. What is a Surrogate Key?

A system-generated key (like an auto-increment ID) used when no natural key exists.

92. What is a Sequence?

A database object that generates a sequence of numeric values, often used for primary keys.

93. What is a Synonym in SQL?

A synonym is an alias for a table, view, or other object, allowing simpler reference.

94. What is the difference between DROP and DELETE in terms of rollback?

`DELETE` can be rolled back if inside a transaction. `DROP` cannot — it removes the structure itself.

95. What is the difference between CHAR and VARCHAR?

`CHAR(n)` has fixed length; `VARCHAR(n)` is variable-length and saves space for shorter strings.

96. What are Aggregate Functions in SQL?

Functions that operate on sets of values and return a single value — `COUNT()`, `AVG()`, `SUM()`, `MIN()`, `MAX()`.

97. What is the difference between GROUP BY and PARTITION BY?

`GROUP BY` aggregates results into single rows per group; `PARTITION BY` divides results into partitions but doesn't collapse rows (used with window functions).

98. What are Window Functions?

They perform calculations across a set of rows related to the current row, like

`ROW_NUMBER()` , `RANK()` , `SUM() OVER()` .

99. What are Recursive Queries?

Queries that refer to themselves to handle hierarchical data — implemented using `WITH RECURSIVE` or Common Table Expressions (CTEs).

100. What is the difference between Logical Delete and Physical Delete?

A logical delete marks a record as inactive (using a flag), while a physical delete removes it from storage.